

Lite Edition

Dickwella Hotel Management System

Data Dictionary

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Legend

-  Primary key
-  Primary key disabled
-  User-defined primary key
-  Unique key
-  Unique key disabled
-  User-defined unique key
-  Active trigger
-  Disabled trigger
-  Many to one relation
-  User-defined many to one relation
-  One to many relation
-  User-defined one to many relation
-  One to one relation
-  User-defined one to one relation
-  Input
-  Output
-  Input/Output
-  Uses dependency
-  User-defined uses dependency
-  Used by dependency
-  User-defined used by dependency

Dickwella Hotel Management System

This technical documentation provides a comprehensive guide on how to use the Hotel Management Database System. The system manages hotel operations, including guest reservations, room assignments, billing, and complaint management. The database stores data across multiple tables, including Guests, Rooms, Reservations, Employees, Additional Services, Bills, and Complaints. This documentation will guide users through the structure and usage of the system.

1. Hotel Management System User Manual

1. Introduction

This Hotel Management System is designed to streamline the management of room bookings, additional services, and billing processes at Dickwella Beach Hotel and Resort. The system keeps track of guest reservations, room availability, employee assignments, and guest complaints, and it generates final bills for stays.

2. System Overview

The system consists of the following main modules:

- **Guest Management**
- **Room Management**
- **Reservation Management**
- **Employee Management**
- **Additional Services Management**
- **Billing**
- **Complaint Management**

Each module interacts with the others to provide a comprehensive management solution for the hotel.

3. Accessing the System

To access the system, users need to have appropriate credentials (e.g., username and password). Once logged in, users can navigate through the system's modules using the menu.

4. Guest Management

4.1 Adding a New Guest

1. Navigate to the **Guest Management** module.
2. Click on **Add New Guest**.
3. Enter the guest's personal details: Name, Date of Birth (DoB), Address, and Age.
4. Click **Save** to store the guest information.

4.2 Updating Guest Information

1. Search for the guest using the **Search** function in the **Guest Management** module.
2. Select the guest and click **Edit**.
3. Make the necessary changes and click **Save**.

4.3 Managing Guest Contacts

1. Access the **Guest Contacts** section under **Guest Management**.
2. Select a guest by their `GuestID`.
3. Add, edit, or delete contact numbers as needed.

5. Room Management

5.1 Viewing Room Availability

1. Go to the **Room Management** module.
2. The **Room List** shows the current status of all rooms (e.g., vacant, occupied).
3. Use the filters to view availability based on room type (Single, Double, Family).

5.2 Adding Room Facilities

1. Navigate to the **Room Facility** section.
2. Select the `RoomID` to which you want to add a facility.
3. Choose the facility (e.g., chair, bed) from the list or enter a custom facility.
4. Click **Add** to save the facility to the room.

6. Reservation Management

6.1 Creating a New Reservation

1. In the **Reservation Management** module, click on **New Reservation**.
2. Enter the guest's `GuestID` or search for the guest.
3. Select the room(s) by `RoomID` and enter the start and end dates of the stay.
4. Specify the number of guests and update the payment status.
5. Click **Save** to finalize the reservation.

6.2 Checking In a Guest

1. Locate the guest's reservation using the **Reservation Management** module.
2. Verify and update guest information as necessary.
3. Mark the reservation as **Checked-In** to register the start of the stay.

6.3 Cancelling a Reservation

1. Search for the reservation by `ReservationID`.
2. Select the reservation and click **Cancel Reservation**.
3. Confirm the cancellation; note that reservation payments are non-refundable.

7. Employee Management

7.1 Assigning Rooms to Employees

1. In the **Employee Management** module, select an employee by `EmployeeID`.
2. Choose the room(s) they are responsible for using `RoomID`.
3. Assign roles (e.g., Room Boy, Waitress) and click **Save**.

7.2 Adding a New Employee

1. Click on **Add New Employee** in the **Employee Management** module.
2. Enter the employee's details, including Name, Role, and Assigned Room.
3. Click **Save** to store the information.

8. Additional Services Management

8.1 Adding a New Service

1. Navigate to the **Additional Services** module.
2. Click on **Add Service**.
3. Enter the service details such as `ServiceName`, `RoomID`, and `ServiceCharge`.
4. Click **Save** to add the service.

8.2 Viewing Services for a Room

1. In the **Additional Services** module, search by `RoomID` to see all services linked to that room.
2. The list will display all services availed during the guest's stay.

9. Billing

9.1 Generating a Final Bill

1. Go to the **Billing** module and search for the guest's `ReservationID`.
2. The system will automatically calculate the total amount, including room charges and additional services.
3. If there are any pre-paid amounts (e.g., reservation payments), the system will deduct these from the final amount.
4. Review the bill details and click **Generate Bill** to create the final bill.

9.2 Viewing Previous Bills

1. Search for a bill using the `BillID` in the **Billing** module.
2. The system will display the full bill history, including total, deducted, and final amounts.

10. Complaint Management

10.1 Logging a Complaint

1. In the **Complaint Management** module, click **Add Complaint**.
2. Enter the `GuestID`, `RoomID`, complaint description, and date.
3. Click **Save** to log the complaint for future reference.

10.2 Resolving a Complaint

1. Locate the complaint using the **Search** function.
2. Update the complaint status to **Resolved** after addressing the issue.
3. Optionally, add a resolution note for reference.

11. Reporting and Analysis

11.1 Generating Reports

1. Access the **Reports** module from the main menu.
2. Select the type of report you want to generate (e.g., room occupancy, service usage, income per room).
3. Set any necessary filters (e.g., date range) and click **Generate**.
4. Reports can be exported to various formats (e.g., PDF, Excel).

12. System Navigation

The system interface includes a navigation bar that allows users to quickly switch between modules. Each module's dashboard provides a summary of current activities and links to detailed functions.

- **Main Menu:** Access modules like Guest Management, Room Management, and Reports.
- **Search Functionality:** Quickly find records using keywords or IDs.
- **Help:** Access system documentation and support resources.

13. Troubleshooting

13.1 Common Issues

- **Login Problems:** Ensure that the username and password are correct. If issues persist, contact the system administrator.
- **Data Not Saving:** Check for required fields and valid input formats.
- **System Slowdown:** Large data operations may cause temporary slowdowns. Wait a moment and try again.

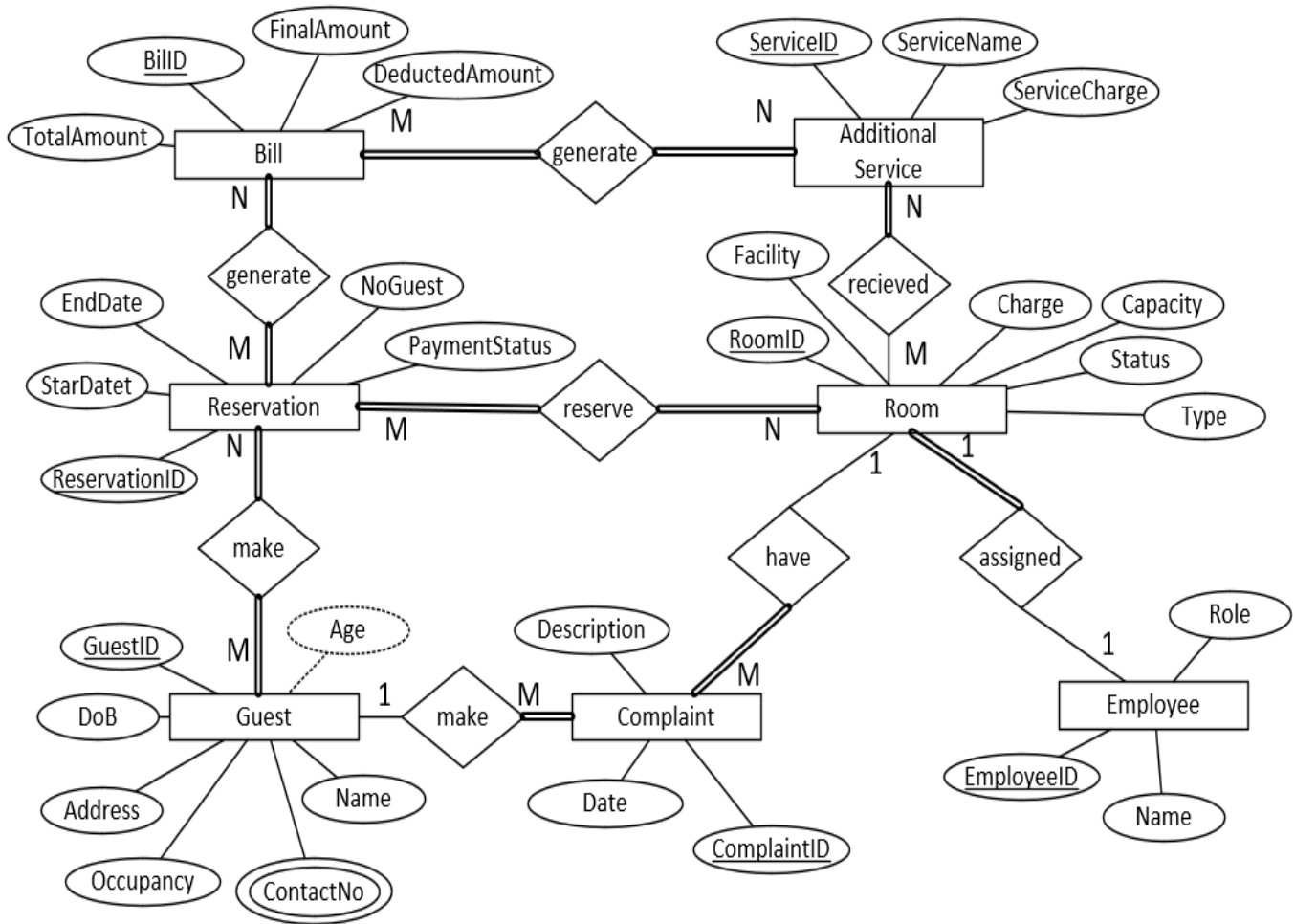
13.2 Support

For additional support, contact the IT department or consult the system's help documentation available under the **Help** menu.

2. Conceptual design of the dickwella hotel management system

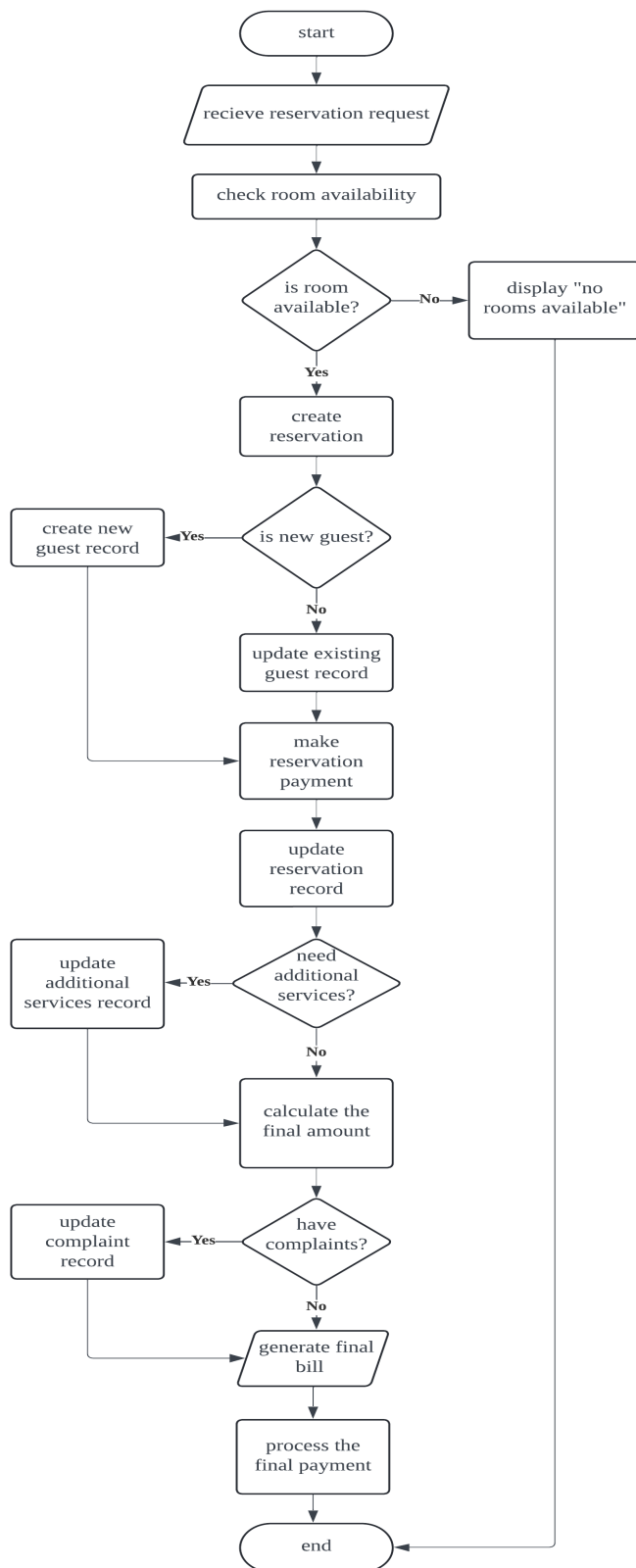
ENTITY	ATTRIBUTES
Guest	<u>GuestID</u> , Name, Occupancy, DoB, Address, Age, ContactNo (multi valued)
Room	<u>RoomID</u> , Facility, Type, Charge, Capacity, Status
Reservation	<u>ReservationID</u> , StartDate, EndDate, PaymentStatus, NoGuest
Employee	<u>EmployeeID</u> , Name, Role, AssignedRoom
Additional Service	<u>ServiceID</u> , ServiceName, ServiceCharge
Bill	<u>BillID</u> , TotalAmount, DeductedAmount, FinalAmount
Complaint	ComplaintID, Date, Description

3. ER diagram for the Dickwella hotel management system



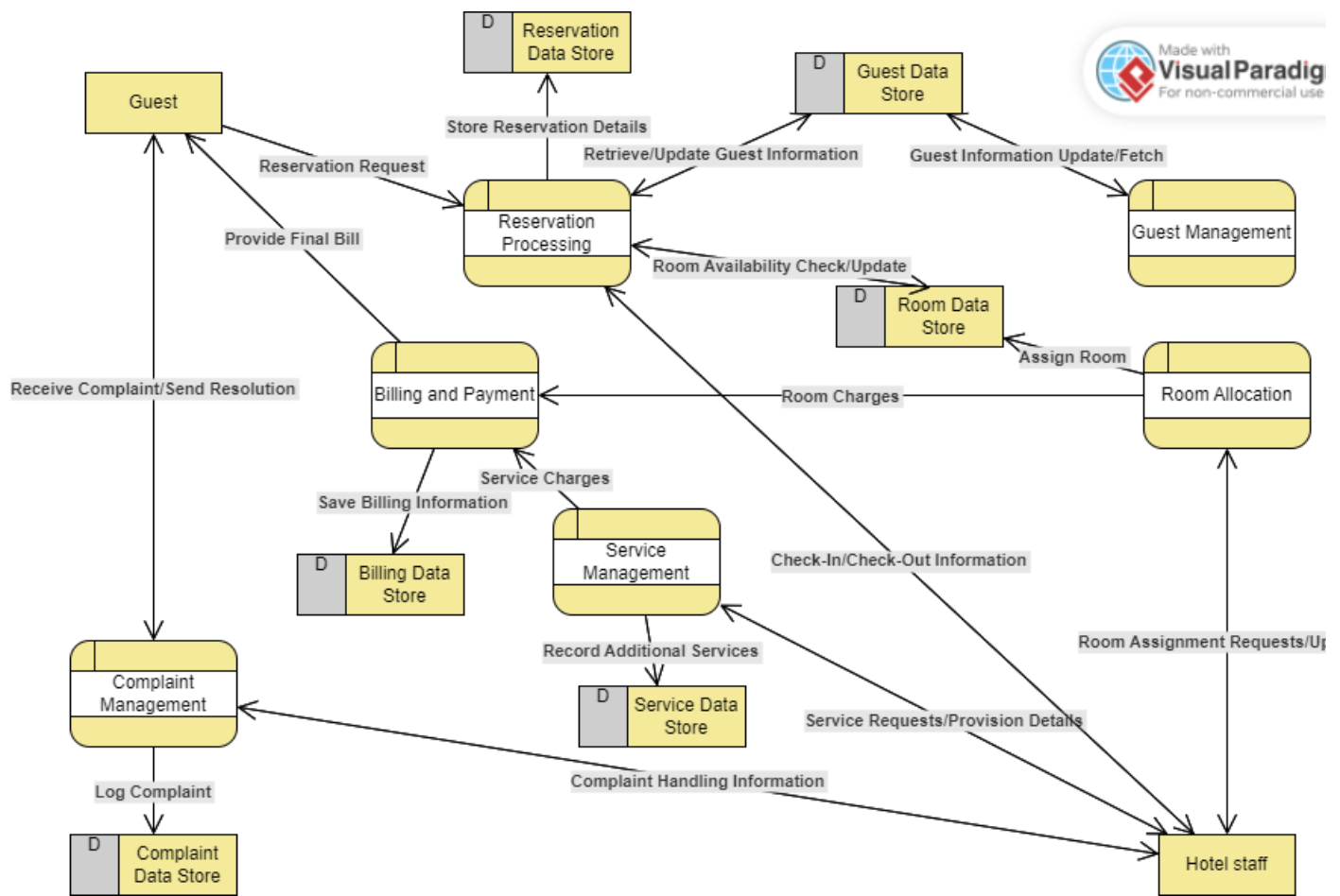
The ER (Entity-Relationship) Diagram provides a detailed view of the data structure within the Dickwella Beach Hotel and Resort's management system.

4. Flow chart of dickwella hotel management system



this flow chart shows the flow of the hotel management system

5. Data flow diagram of Dickwella hotel management system



Overview:

The DFD depicts the flow of information between the **Guest**, **Hotel Staff**, and various internal processes such as **Reservation Processing**, **Room Allocation**, **Service Management**, **Billing and Payment**, and **Complaint Management**. The interactions are supported by data storage areas where vital information is retained for smooth operation and decision-making.

Key Components and Data Flows:

1. Guest Interaction:

- The diagram begins with the **Guest** entity, who initiates the process by making a **Reservation Request**. This data is sent to the **Reservation Processing** process.
- Guests also interact with the system by receiving their **Final Bill** from the **Billing and Payment** process and lodging **Complaints** to the **Complaint Management** process.

2. Reservation Processing:

- **Reservation Processing** is a central process in the system. It receives reservation requests from guests and checks room availability by interacting with the **Room Data Store**.
- It also manages guest details, retrieving or updating information from the **Guest Data Store** and storing final reservation details in the **Reservation Data Store**.
- During check-in and check-out, **Hotel Staff** interact with this process to ensure the correct processing of guest stays.

3. Room Allocation:

- **Room Allocation** assigns rooms to guests based on availability. This process interacts directly with the **Room Data Store** to update room status and coordinates with **Hotel Staff** for room assignments.

4. Service Management:

- **Service Management** is responsible for recording any additional services or products guests purchase during their stay. These details are stored in the **Service Data Store**.
- **Hotel Staff** interact with this process to manage service requests and track service provisions.

5. Billing and Payment:

- This process calculates the final bill for guests based on their reservations and additional services consumed. The calculated bills are stored in the **Billing Data Store**, and the final bill is provided to the guest upon checkout.

5. Complaint Management:

- **Complaint Management** handles any dissatisfaction or issues raised by guests. It records these complaints in the **Complaint Data Store** and facilitates resolution through communication with **Hotel Staff**.

7. Hotel Staff Interaction:

- **Hotel Staff** play a critical role in the daily operations of the hotel, interacting with various processes. They are involved in room assignments, service management, complaint handling, and check-in/check-out procedures, ensuring that guest needs are met efficiently.

Data Flow Types:

- **Bidirectional Data Flows:** These flows represent processes where information is exchanged back and forth between the entities and processes, such as between **Hotel Staff** and **Room Allocation** or **Service Management**.
- **Unidirectional Data Flows:** These flows depict data moving in a single direction, such as a **Reservation Request** from the **Guest** to **Reservation Processing** or **Complaint Logging** from **Complaint Management** to the **Complaint Data Store**.

Conclusion:

The DFD for the hotel management system outlines the efficient movement of data, ensuring that all interactions—from reservations to billing, service management, and complaint handling—are smoothly integrated. The system supports both the operational needs of the hotel staff and the service expectations of the guests, providing a comprehensive solution for managing hotel operations.

6. Other

6.1. Tables

6.1.1. Table: `dbo.AdditionalServices`

- **Description:** Records additional services provided to guests, such as room service or spa treatments, along with associated charges.
- **Purpose of the table:** To track additional services availed by guests, ensuring accurate billing and service management.
- **Linked Tables:**
 - **Room:** Services are associated with specific rooms using `RoomID`.
 - **Bill:** The charges for these services are included in the final bill via `ServiceID`.
- **Columns:** `ServiceID` (Primary Key), `RoomID` (Foreign Key), `ServiceName`, `ServiceCharge`.

Columns

	Name	Data type	Description / Attributes
	<code>ServiceID</code>	<code>varchar(10)</code>	
	<code>RoomID</code>	<code>varchar(10)</code>	Nullable References: <code>dbo.Room</code>
	<code>ServiceName</code>	<code>varchar(50)</code>	Nullable
	<code>ServiceCharge</code>	<code>decimal(10, 2)</code>	Nullable

6.1.2. Table: `dbo.Bill`

- **Description:** Consolidates all financial transactions related to a guest's stay, including room charges, additional services, and deductions.
- **Purpose of the table:** To generate and manage the final bill for each guest, summarizing all charges and payments.
- **Linked Tables:**
 - **Reservation:** The final bill is linked to the reservation via `ReservationID`.
 - **Additional Services:** Charges for additional services are included via `ServiceID`.
- **Columns:** `BillID` (Primary Key), `ServiceID` (Foreign Key), `ReservationID` (Foreign Key), `TotalAmount`, `DeductedAmount`, `FinalAmount`.

Columns

	Name	Data type	Description / Attributes
	<code>BillID</code>	<code>varchar(10)</code>	
	<code>ServiceID</code>	<code>varchar(10)</code>	Nullable References: <code>dbo.AdditionalServices</code>
	<code>ReservationID</code>	<code>varchar(10)</code>	Nullable References: <code>dbo.Reservation</code>
	<code>TotalAmount</code>	<code>decimal(10, 2)</code>	Nullable
	<code>DeductedAmount</code>	<code>decimal(10, 2)</code>	Nullable
	<code>FinalAmount</code>	<code>decimal(10, 2)</code>	Nullable

6.1.3. Table: `dbo.Complaints`

- **Description:** Logs complaints made by guests regarding rooms or services, including details of the issue and the date of the complaint.
- **Purpose of the table:** To document and manage guest complaints, facilitating follow-up actions to resolve issues and improve service quality.
- **Linked Tables:**
 - **Guest:** Complaints are linked to the guest who made them via `GuestID`.
 - **Room:** The room associated with the complaint is linked through `RoomID`.
- **Columns:** `ComplaintID` (Primary Key), `GuestID` (Primary Key), `RoomID` (Foreign Key), `Description`, `Date`.

Columns

	Name	Data type	Description / Attributes
	<code>ComplaintID</code>	<code>varchar(10)</code>	
	<code>GuestID</code>	<code>varchar(10)</code>	Nullable References: <code>dbo.Guest</code>
	<code>RoomID</code>	<code>varchar(10)</code>	Nullable References: <code>dbo.Room</code>
	<code>Description</code>	<code>varchar(255)</code>	Nullable
	<code>Date</code>	<code>date</code>	Nullable

6.1.4. Table: `dbo.Employee`

- **Description:** Stores information about hotel employees, including their roles and room assignments.
- **Purpose of the table:** To manage employee information and their assigned tasks or rooms within the hotel.
- **Linked Tables:**
 - **Room:** Employees are assigned to specific rooms via `RoomID`.
- **Columns:** `EmployeeID` (Primary Key), `RoomID` (Foreign Key), `Name`, `Role`, `AssignedRoom`.

Columns

	Name	Data type	Description / Attributes
	<code>EmployeeID</code>	<code>varchar(10)</code>	
	<code>RoomID</code>	<code>varchar(10)</code>	Nullable References: <code>dbo.Room</code>
	<code>Name</code>	<code>varchar(50)</code>	Nullable
	<code>Role</code>	<code>varchar(30)</code>	Nullable
	<code>AssignedRoom</code>	<code>varchar(10)</code>	Nullable

6.1.5. Table: dbo.Guest

- **Description:** Stores detailed personal and demographic information about guests, including their occupancy status and contact information.
- **Purpose of the table:** To maintain a central repository of guest information, facilitating reservations, billing, and personalized services.
- **Linked Tables:**
 - **Reservation:** Each guest's stay is tracked via the `Reservation` table using `GuestID`.
 - **Complaint:** Complaints made by guests are linked via `GuestID`.
 - **Guest Contacts:** Multiple contact numbers for guests are stored in a separate table linked by `GuestID`.
- **Columns:** `GuestID` (Primary Key), `Name`, `Occupancy`, `DoB`, `Address`, `Age`.

Columns

	Name	Data type	Description / Attributes
🔑	GuestID	varchar(10)	
	Name	varchar(50)	Nullable
	Occupancy	varchar(10)	Nullable
	DoB	date	Nullable
	Address	varchar(100)	Nullable
	Age	int	Nullable

6.1.6. Table: dbo.GuestContacts

- **Description:** Stores contact numbers associated with each guest.
- **Purpose of the table:** To maintain a list of contact numbers for each guest, allowing for multiple points of contact.
- **Linked Tables:**
 - **Guest:** Linked via `GuestID` to identify which guest each contact number belongs to.
- **Columns:** `GuestID` (Primary Key), `ContactNo` (Primary Key).

Columns

	Name	Data type	Description / Attributes
🔑	GuestID	varchar(10)	References: <code>dbo.Guest</code>
🔑	ContactNo	varchar(15)	

6.1.7. Table: dbo.Reservation

- **Description:** Manages guest reservations, including the details of the guest, room, and payment status.
- **Purpose of the table:** To record and track reservations, ensuring accurate room assignments and billing.
- **Linked Tables:**
 - **Guest:** Reservations are linked to guests using `GuestID`.
 - **Room:** Rooms booked are linked through `RoomID`.
 - **Bill:** The final billing for the reservation is linked via `ReservationID`.
- **Columns:** `ReservationID` (Primary Key), `GuestID` (Foreign Key), `RoomID` (Foreign Key), `StartDate`, `EndDate`, `NoGuest`, `PaymentStatus`.

Columns

	Name	Data type	Description / Attributes
	ReservationID	varchar(10)	
	GuestID	varchar(10)	Nullable References: dbo.Guest
	RoomID	varchar(10)	Nullable References: dbo.Room
	StartDate	date	Nullable
	EndDate	date	Nullable
	NoGuest	int	Nullable
	PaymentStatus	varchar(20)	Nullable

6.1.8. Table: dbo.Room

- **Description:** Records details about each room in the hotel, including type and availability status.
- **Purpose of the table:** To manage room inventory, track availability, and support the reservation process.
- **Linked Tables:**
 - **Reservation:** Rooms are booked through the `Reservation` table using `RoomID`.
 - **Room Facility:** Facilities associated with each room are stored in a separate table linked by `RoomID`.
 - **Employee:** Employees such as room boys and waiters are assigned to rooms through `RoomID`.
- **Columns:** `RoomID` (Primary Key), `Type`, `Status`.

Columns

	Name	Data type	Description / Attributes
	RoomID	varchar(10)	
	Type	varchar(20)	Nullable
	Status	varchar(20)	Nullable

6.1.9. Table: dbo.RoomDetail

- **Description:** Stores pricing and capacity information for different room types.
- **Purpose of the table:** To maintain standard charges and capacity limits for each room type, supporting the booking and billing processes.
- **Linked Tables:**
 - **Room:** Room type details are used in conjunction with the `Room` table.
- **Columns:** Type (Primary Key), Charge, Capacity.

Columns

	Name	Data type	Description / Attributes
🔑	Type	varchar(20)	
	Charge	decimal(10, 2)	Nullable
	Capacity	int	Nullable

6.1.10. Table: dbo.RoomFacility

- **Description:** Contains details about the facilities available in each room, such as furniture and amenities.
- **Purpose of the table:** To track the facilities associated with each room, aiding in room management and guest services.
- **Linked Tables:**
 - **Room:** Facilities are linked to specific rooms using `RoomID`.
- **Columns:** RoomID (Primary Key), Facility (Primary Key).

Columns

	Name	Data type	Description / Attributes
🔑	RoomID	varchar(10)	
🔑	Facility	varchar(50)	